

Today, we will again be working with the Palmer penguins data frame.

Question 1

For this first question, **do not open RStudio**. Consider the relationship between bill length and bill depth when broken down by species (Adelie, Chinstrap, Gentoo). Here are some summary statistics on this information, for reference.

```
# A tibble: 3 × 5
  species `Mean Bill Length` `SD Bill Length` `Mean Bill Depth` `SD Bill Depth`
  <fct>          <dbl>          <dbl>          <dbl>          <dbl>
1 Adelie          38.8            2.66            18.3            1.22
2 Chinstrap       48.8            3.34            18.4            1.14
3 Gentoo          47.6            3.11            15.0            0.986
```

part a

Sketch a plot which can be used to summarize the relationship between the three variables. *Depict a shape which reflects your expectation of the phenomenon.* Label your axes and give the plot a title.

part b

State the aesthetic mappings and geometry involved in the plot.

Using the penguins data frame, answer the following questions. Where code is needed, you may write it down in the space given.

Question 2

part a

Use `lm` to fit a linear model to estimate bill depth using both bill length and species. Write the code you used below.

Question 3

Bella the penguin has a bill length of 43 millimeters. Using the linear model output, estimate her bill depth (in millimeters) in three different cases:

part a

The species of the penguin is Adelie.

part b

The species of the penguin is Chinstrap.

part c

The species of the penguin is Gentoo.

Question 4

Fit the multiple linear regression model from **Question 2**, but instead having "Chinstrap" as the reference level in the species variable. Copy the code you used to make the change of reference level (you do not need to copy the linear model code).